

JAVIER AHMAD

Project Engineer — SCADA, Instrumentation & Industrial Automation

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PROFESSIONAL SUMMARY

Project Engineer who takes industrial monitoring systems from sensor to server, commissioning SCADA/HMI and instrumentation for clients including PLN Indonesia Power, Pelindo, PT Pertamina. Has integrated 500+ field devices via Modbus TCP/RTU, DNP3, OPC UA, and MQTT, cut installation time by 20%+, and led root cause analysis to restore data visibility on live sites. Internal Auditor for ISO 9001, 14001, and 45001.

CORE SKILLS

Technical Skills: Python, HMI (Weintek), PLC Allen-Bradley, PLC Siemens, PLC Mitsubishi, Modbus Protocol, OPC UA, MQTT, Ladder Logic Programming, DCS/PLC Familiarity, Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT)

Instrumentation & SCADA Engineering: Instrumentation Commissioning, SCADA/HMI Configuration & Troubleshooting, I&E Field Supervision, Root Cause Analysis (RCA), Sensor & Instrument Calibration, Modbus/OPC UA/DNP3 Communication

Project Delivery & Compliance: Project Schedule Tracking, Risk Assessment, HSE Compliance, Technical Documentation, Internal Auditing (ISO 9001/14001/45001)

WORK EXPERIENCE

Project Engineer

Feb 2025 – Present

PT Daun Biru Engineering | Depok, West Java, Indonesia

- Configure and commission end-to-end industrial monitoring systems, sensors, RTUs, PLC, and HMI — to server, via Modbus TCP/RTU, DNP3, OPC UA, MQTT, and GET/POST HTTPS.
- Coordinate Instrument & Electrical (I&E) technicians during site installations, optimizing workflows to improve installation time by 20% while maintaining 100% HSE safety compliance.
- Lead root cause analysis (RCA) on SCADA and vibration monitoring systems, resolving hardware and communication issues to restore data visibility on live client sites.
- Serve as Internal Auditor for ISO 9001, 14001, and 45001, ensuring all instrumentation deployments align with global QHSE standards.

Key Projects Delivered

- **ITB (Chemistry Laboratory)** — Project Manager for a Furnace Monitoring System pilot, integrating 30+ temperature sensor points, full panel wiring, and a Weintek HMI with trending, delivered in 14 days with 100% functional commissioning and zero rework.
- **PLN Indonesia Power** — Installed and configured the MONITA online monitoring system across 11 diesel engines at PLTD Selayar, bringing 300+ measurement points online via mixed protocols (OPC UA, Modbus RTU/TCP, DNP3) to a central server.
- **Pelindo** — Built a SCADA/HMI system for Tanjung Mas Port covering 12 polder pumps and 6 water basins on a single Mitsubishi FX3U PLC, achieving alarm response times under 3 seconds via one integrated web dashboard.
- **Hotel Ascott** — Configured Weintek HMI and Modbus RTU monitoring for 4 heat pump units, reaching 98.5% uptime and cutting manual inspection time by ~10 hours/month through automated reporting.
- **PT Hutama Karya (Persero)** — Troubleshot a Rockwell/Allen-Bradley Emonitor CMS vibration monitoring system, diagnosing faults across multiple panel blocks and coordinating punchlist items directly with Rockwell.
- **PTPN 1** — Diagnosed and resolved MONITA module failures halting data flow, restoring data transmission with full calibration certification handover.

Intern, Department of Distribution and Lifting

Mar 2024 – Aug 2024

SKK MIGAS | Jakarta, Indonesia

- Supported the coordination team in monitoring a 4,000+ BOPD production decline recovery at PT Bumi Siak Pusako, tracking pipeline leak restoration progress that returned output to 6,500+ BOPD.
- Evaluated technical requirement updates for custody transfer metering devices (crude oil, gas orifice, ultrasonic, and prover/master meters) against national upstream standards.
- Monitored oil and gas sales activities across 106 contractors (KKKS) under the Production Sharing Contract (PSC) system.

- Participated in Facility Acceptance Tests (FAT) and documented Facility Sharing Agreements between contractors to ensure regulatory alignment.

Intern, Electrical and Instrumentation Engineer

Jan 2023 – Feb 2023

PT Pertamina (Persero) Refinery Unit VI Balongan | Indramayu, West Java, Indonesia

- Interpreted and identified 10+ P&ID and PFD diagrams for the Unit 32-F-104 Debutanizer Reboiler Heater (naphtha processing), ensuring accuracy in instrument-to-process mapping.
- Evaluated safeguard specifications for heater units by analyzing 3 nodes within HAZOP documents, identifying potential hazards and verifying the reliability of existing instrumentation safeguards.
- Gained technical exposure to refinery-scale instrumentation, including flow meters, pressure transmitters, level sensors, and DCS/PLC architectures.

CERTIFICATIONS

- Internal Auditor — ISO 9001 (Quality), ISO 14001 (Environmental), ISO 45001 (Occupational Health & Safety) (PT SEGORO INTERKON)
- Senior Executor, Construction and Installation of Electrical Equipment, competent to coordinate diesel power plant (PLTD) development and install control and instrumentation systems (DCIS), 2025 (PT SERTIFIKASI TEKNIK LISTRIK)

EDUCATION

Bachelor of Engineering Physics

Sep 2020 – Oct 2024

Universitas Gadjah Mada | Yogyakarta, Indonesia

- Relevant Coursework: Measurement & Monitoring, Instrumentation Systems, Sensor Technology, Actuator Technology, Control Engineering, PLC Application, Automation System Design, Safety Instrumented System
- GPA: 3.35 / 4.00
- Outstanding Student, Engineering Physics Program, 2021

ACHIEVEMENTS

- 1st Place, Olgemas Paper Competition (Faculty of Geography, UGM), 2022 — “Utilization of Syngas in Integrated Solar Combined Cycle Power Using Former Mining Lands to Address Energy Inequality in Kalimantan, Indonesia.”
- CAPTURE (Carbon Abatement Performance Tracking and Utilization with Real-time Evaluation) — Incentive award, Student Creativity Week, Ministry of Education and Culture, 2023. Designed a real-time IoT monitoring architecture and sensor firmware for carbon capture emission tracking.
- 2nd Place, Evergreen Essay Competition (FKM UI), 2021 — Essay on biodegradable masks.

ORGANIZATIONAL EXPERIENCE

Event Chair (PT Daun Biru Engineering Representative)

Nov 2025, Aug 2026

"Vibration Analysis on Rotating Machinery" Training — BKM-PII x PT Daun Biru Engineering

- Led project management (timeline, S-curve tracking) and coordinated instructors and participants for a 4-day ISO 18436 Level 2-equivalent certification training, benchmarked against Mobius Institute standards.
- Positioned the training as the first of its kind delivered in partnership with Persatuan Insinyur Indonesia (PII) as legal stakeholder and Majelis Uji Kompetensi (MUK) as examination body.
- Ran the program twice (Nov 2025, Aug 2026), achieving 100% completion among all 18 participants across both cohorts.